Solvay Brussels School
Economics & Management

GEST-S2701 – Exercise Session

Cash flows + Working capital requirement (WCR)

2025-2026

EQUIS ACCREDITED
Association of MBAs
ULB

ACTIF	PASSIF
ΔFE	ΔCAP
	ΔRS
	ΔRES
	ΔRE
ΔFA	
	ΔPROV
	ΔSUBS
	ΔLTDfin + ΔSTDfin
ΔARfin	
ΔWCR	
ΔOS	
ΔCASH	

Colours indicate the restated balance sheet items related to:

- **CF from operating activities (CF_{op})**
- **CF from investing activities (CF_{inv})**
- **CF from financing activities (CF_{fin})**
- **ΔCASH**

Exercise 2.1

For each of the operations below, identify the direct impacts on cash flows by distinguishing between the different types of cash flows.

Assumption: unless otherwise mentioned, there is no VAT.

Exercise 2.1

#	Operation	Timing	CF_{op} (1)	CF_{inv} (2)	FCF (3) = (1)+(2)	CF_{fin} (4)	Δ CASH (5) = (3)+(4)
1	Purchase on credit of goods	At the time of payment of the goods (which does not necessarily correspond to the time of purchase!)	-	0	-	0	-
2	Repayment of a loan from a credit institution	At the time of repayment	0	0	0	-	-
3	Purchase of a production machine to increase efficiency	At the time of cash payment	0	-	-	0	-
4	Sale of a production factory	At the time of receipt of payment	0	+	+	0	+
5	Payment of corporate tax to the State	At the time of payment	-	0	-	0	-
6	Receipt of a VAT refund from the administration	At the time of receipt of payment	+	0	+	0	+
7	Depreciation of a production machine	At the time of depreciation	+	0	+	-	0
8	Settlement of an invoice for major repair	At the time of payment	-	0	-	0	-
9	Loan from a shareholder	At the time of receipt of the borrowed amount	0	0	0	+	+
10	Receipt of an electricity invoice from a supplier	At the time of receipt of the invoice	0	0	0	0	0
		At the time of payment of the invoice	-	0	-	0	-

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Exercise 2.1

#	Operation	Timing	CF_{op} (1)	CF_{inv} (2)	FCF (3) = (1)+(2)	CF_{fin} (4)	Δ CASH (5) = (3)+(4)
11	Services provided to a client, payable "30 days end of month"	At the time of the sending of the invoice	0	0	0	0	0
		At the time of payment	+	0	+	0	+
12	Payment of an invoice for legal fees to a law firm	At the time of payment	-	0	-	0	-
13	Payment of rent for offices	At the time of payment	-	0	-	0	-
14	Purchase of a car intended for resale (self-financed by the company)	At the time of payment	-	0	-	0	-
15	Purchase of a company car (self-financed by the company)	At the time of payment	0	-	-	0	-
16	Purchase of a company car (financed by credit granted by the financial arm of the car dealer)	At the time of purchase (by assumption: no initial disbursement of the vehicle amount as credit is granted concurrently)	0	-	-	+	0
17	Long-term equity investment in a company	At the time of payment of shares	0	-	-	0	-
18	Purchase of shares (short-term) to use available liquidity	At the time of payment of shares	0	0	0	0	0
19	Payment by the bank of interest on the bank account	At the time of receipt of payment	+	0	+	0	+
20	Payment by the company of interest to the bank	At the time of payment	-	0	-	0	-

Exercise 2.2

The company ABC was created at the end of 2024. It closed the year with a receivable of 10,000€ and an amount in the "Available values" account equal to 20,000€. During the 2025 fiscal year, there are no investment or financing cash flows (in this case, the change in cash is always equal to the operating cash flow).

1) For each of the operations below, calculate:

- The incremental CFop (= due to the operation) according to the direct method
- The incremental net result (= due to the operation)
- The incremental WCR (= due to the operation)
- The incremental CFop (= due to the operation) according to the indirect method

By assumption, there is no VAT.

Exercise 2.2

0. On 01/01, the company proceeds with the opening of its accounts.

# Op	0	1	2	3	4	5	6	7	8
NR	0								
WCR	10,000								
ΔWCR	0								
CFop	0								

Exercise 2.2

1. On 15/01, it invoices its services to company XYZ for an amount of 15,000€. The invoice is payable at 30 days end of month.

# Op	0	1	2	3	4	5	6	7	8
NR	0	15,000							
WCR	10,000	25,000							
ΔWCR	0	15,000							
CFop	0	0							

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Exercise 2.2

2. On 01/02, it receives payment related to the receivable opened in 2024 (valued at 10,000€).

# Op	0	1	2	3	4	5	6	7	8
NR	0	15,000	15,000						
WCR	10,000	25,000	15,000						
ΔWCR	0	15,000	5,000						
CFop	0	0	10,000						

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Exercise 2.2

3. On 05/02, it receives an invoice related to the services of a consulting company. This invoice amounts to 20,000€ and ABC pays directly.

# Op	0	1	2	3	4	5	6	7	8
NR	0	15,000	15,000	-5,000					
WCR	10,000	25,000	15,000	15,000					
ΔWCR	0	15,000	5,000	5,000					
CFop	0	0	10,000	-10,000					

Exercise 2.2

4. On 10/02, it receives an invoice from its lawyer, amounting to 15,000€. The invoice is payable within two weeks.

# Op	0	1	2	3	4	5	6	7	8
NR	0	15,000	15,000	-5,000	-20,000				
WCR	10,000	25,000	15,000	15,000	0				
ΔWCR	0	15,000	5,000	5,000	-10,000				
CFop	0	0	10,000	-10,000	-10,000				

Exercise 2.2

5. On 15/02, it receives from company XYZ the payment for the services invoiced on 15/01 (15,000€).

# Op	0	1	2	3	4	5	6	7	8
NR	0	15,000	15,000	-5,000	-20,000	-20,000			
WCR	10,000	25,000	15,000	15,000	0	-15,000			
ΔWCR	0	15,000	5,000	5,000	-10,000	-25,000			
CFop	0	0	10,000	-10,000	-10,000	5,000			

Exercise 2.2

6. On 20/02, it settles in cash, its invoice related to lawyer's fees (15,000€).

# Op	0	1	2	3	4	5	6	7	8
NR	0	15,000	15,000	-5,000	-20,000	-20,000	-20,000		
WCR	10,000	25,000	15,000	15,000	0	-15,000	0		
ΔWCR	0	15,000	5,000	5,000	-10,000	-25,000	-10,000		
CFop	0	0	10,000	-10,000	-10,000	5,000	-10,000		

Exercise 2.2

7. On 01/03, it invoices its services to company UVW for an amount of 30,000€. The client pays directly.

# Op	0	1	2	3	4	5	6	7	8
NR	0	15,000	15,000	-5,000	-20,000	-20,000	-20,000	10,000	
WCR	10,000	25,000	15,000	15,000	0	-15,000	0	0	
ΔWCR	0	15,000	5,000	5,000	-10,000	-25,000	-10,000	-10,000	
CFop	0	0	10,000	-10,000	-10,000	5,000	-10,000	20,000	

Exercise 2.2

8. On 15/03, it receives from its energy supplier an invoice amounting to 10,000€. The invoice is payable at 60 days end of month.

# Op	0	1	2	3	4	5	6	7	8
NR	0	15,000	15,000	-5,000	-20,000	-20,000	-20,000	10,000	0
WCR	10,000	25,000	15,000	15,000	0	-15,000	0	0	-10,000
ΔWCR	0	15,000	5,000	5,000	-10,000	-25,000	-10,000	-10,000	-20,000
CFop	0	0	10,000	-10,000	-10,000	5,000	-10,000	20,000	20,000

Exercise 2.2

2) For the entire fiscal year (i.e., for operations 1 to 8), calculate:

- The CFop according to the direct method

The CFop is the sum of the impacts of the different operations on CFop:

= 0 (#1) + 10.000 (#2) -20.000 (#3) + 0 (#4) + 15.000 (#5) – 15.000 (#6) + 30.000 (#7) + 0 (#8)

= + 20.000 €

Exercise 2.2

2) For the entire fiscal year (i.e., for operations 1 to 8), calculate:

- The CFop according to the direct method
- The net result

The net result is the sum of revenues – the sum of expenses for the year = 0€

# Op	0	1	2	3	4	5	6	7	8	Cum. (1-8)	
NR incr.	0	+15,000	0	-20,000	-15,000	0	0	+30,000	-10,000	0	= NR
WCR incr.	0	+15,000	-10,000	0	-15,000	-15,000	+15,000	0	-10,000	-20,000	
CFop incr.	0	0	+10,000	-20,000	0	+15,000	-15,000	+30,000	0	+20,000	

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Exercise 2.2

2) For the entire fiscal year (i.e., for operations 1 to 8), calculate:

- The CFop according to the direct method
- The net result
- The variation in WCR. Why do we consider the variation in WCR, and not the WCR alone?

The variation in WCR is the WCR at year-end – WCR at year-start = -10.000 – 10.000 = -20.000 €

# Op	0	1	2	3	4	5	6	7	8	Cum. (1-8)	
NR incr.	0	+15,000	0	-20,000	-15,000	0	0	+30,000	-10,000	0	
WCR incr.	0	+15,000	-10,000	0	-15,000	-15,000	+15,000	0	-10,000	-20,000	= ΔWCR
CFop incr.	0	0	+10,000	-20,000	0	+15,000	-15,000	+30,000	0	+20,000	

Exercise 2.2

2) For the entire fiscal year (i.e., for operations 1 to 8), calculate:

- The CFop according to the direct method
- The net result
- The variation in WCR. Why do we consider the variation in WCR, and not the WCR alone?

The net result and the CFop are flows, while the WCR is a stock. To obtain the CFop (i.e., the change in operating cash since the beginning of the year), it is necessary to subtract from the net result (i.e., the change in revenues and expenses since the beginning of the year) the variation in WCR since the beginning of the year.

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Exercise 2.2

2) For the entire fiscal year (i.e., for operations 1 to 8), calculate:
- The CFop according to the direct method
- The net result
- The variation in WCR. Why do we consider the variation in WCR, and not the WCR alone?
- The CFop according to the indirect method

The CFop is obtained, according to the indirect method, by subtracting from the net result the variation in WCR
= 0 – (-20,000) = + 20,000 €

# Op	0	1	2	3	4	5	6	7	8	Cum. (1-8)	
NR incr.	0	+15,000	0	-20,000	-15,000	0	0	+30,000	-10,000	0	
WCR incr.	0	+15,000	-10,000	0	-15,000	-15,000	+15,000	0	-10,000	-20,000	
CFop incr.	0	0	+10,000	-20,000	0	+15,000	-15,000	+30,000	0	+20,000	= CFop

Exercise 2.2

3) How do you explain that at the end of the different operations, the CFop is positive while the net result is zero?

The CFop differs from the net result for 2 reasons:

- It takes into account the receipt of the receivable opened of 10,000€ in 2024 (Op 2), which appears in the net result of 2024 and not in the net result of 2025..
- It does not yet take into account the energy invoice of 10,000€, since it has not yet been paid.

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Exercise 2.3

A company presents the following balance sheets (in thousands of euros) in 2022 and 2023. In 2023, it recorded depreciations amounting to 300k euros and achieved a net profit of 0 €.

BS at 31/12/2022

6,000	**ASSET**	**LIABILITY**	**6,000**
	FE	CAP	800
		RS	-
		RES	-
200	FA	RE	-
		SUBS	-
		PRDT	200
		LTD	-
3,500	Inventory		
1,800	AR	AP	5,000
-	ACC(A)	ACC(L)	-
-	INVST		
500	CASH		

BS at 31/12/2023

6,500	**ASSET**	**LIABILITY**	**6,500**
	FE	CAP	1,800
		RS	-
		RES	-
1,000	FA	RE	-
		SUBS	-
		PRDT	200
		LTD	3,000
1,500	Inventory		
1,000	AR	AP	1,500
-	ACC(A)	ACC(L)	-
-	INVST		
3,000	CASH		

Exercise 2.3 – 1) MCQ

1) The cash position of the company have considerably increased between the two fiscal years. What is/are the reason(s) that contribute(s) to this increase?

Exercise 2.3 – 1) MCQ

a) The acquisition of new depreciable assets for a value of 1,100 €.

FALSE : an acquisition (investment) primarily decreases cash (negative investment cash flow).

b) A new loan of 3,000 € from a credit institution.

TRUE : a new loan increases cash (positive financing cash flow) and in the presented balance sheets, 3,000 € corresponds to the increase in financial debts during the period.

c) A capital increase of 1,000 €, paid in cash.

TRUE : a capital increase increases cash (positive financing cash flow) and in the presented balance sheets, 1,000 € corresponds to the increase in contribution during the period.

d) A decrease in the AP.

FALSE : a decrease in operating debts increases WCR and decreases cash (negative operating cash flow).

e) A decrease in the AR.

TRUE : a decrease in customer receivables decreases WCR and increases cash (positive operating cash flow).

Exercise 2.3 – 2) WCR

2) Calculate the working capital requirement (WCR) at the end of 2022 and end of 2023. 2 Comment on the impact of WCR on the company's cash position.

Exercise 2.3 – 2) BFR

- **WCR = Inventory + AR –AP + Acc(A) – Acc(L)**

- 31/12/2022 : WCR = 3,500 + 1,800 – 5,000 + 0 – 0 = 300 €

- 31/12/2023 : WCR = 1,500 + 1,000 – 1,500 + 0 – 0 = 1,000 €

- Variation in WCR over the period: Δ WCR = 1,000 – 300 = 700 €

- Increase in WCR → Negative impact on the company's (operating) cash flow

Exercise 2.3 – 3) CF statement (indirect method)

3) Prepare the cash flow statement ("indirect method"), distinguishing between operating, investment, and financing cash flows.

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Exercise 2.3 – 3) CF statement (méthode indirecte)

Operating CF (CF op) =	**– 400 €**
Net profit	0
+ Δ PROV	+ 0 – 0
– Δ WCR	– 700
+ Depreciation_{FA}	+ 300
Investment CF (CF inv) =	**– 1,100 €**
– Δ (FE + FA)	– 800
– Depreciation_{FA}	– 300
Financing CF (CF fin) =	**4,000 €**
Δ CAP	1,000
+ Δ Debts – Δ Financials credits	+ 3,000
CF op + CF inv + CF fin =	**2,500 €**

Exercise 2.3 – 4) Verification of CF

4) How can you verify your answer?

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Exercise 2.3 – 4) Verification of CF

- **CF op + CF inv + CF fin = Δ CASH**

- CF op + CF inv + CF fin = 2,500 € (cf CF statement)

- Δ CASH = CASH (31/12/2023) – CASH (31/12/2022) = 3,000 – 500 = 2,500 € (cf beginning and end of period balance sheets)

- OK